Digital Roadmaps

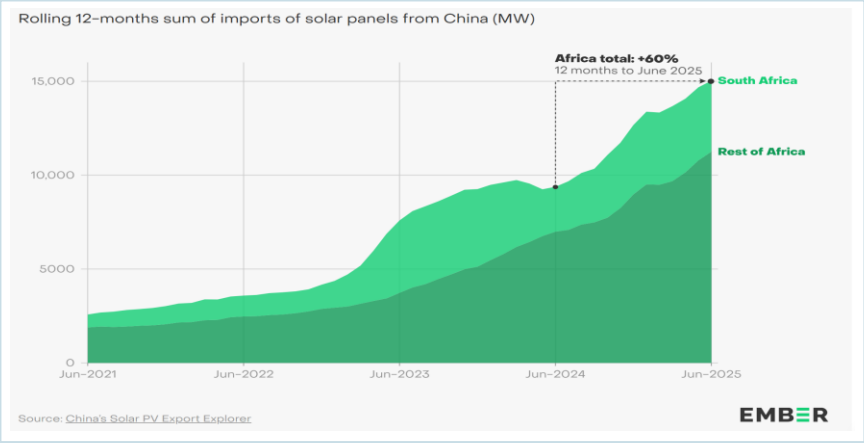
Snapshot from sub-Saharan Africa

Kabir Malik · ESMAP / World Bank Group

ISUW, 2026

The Digital Imperative in Africa

Legacy challenges are colliding with the needs of the future power system



60%

Y-o-Y solar panel import surge
Record 15,032 MW · 12 months to Jun 2025 · EMBER

20,000+

EV charging points by 2025
Up from <1,000 in 2021 · ev24.africa

~19%

T&D losses — West Africa
vs. 5.9% OECD benchmark · UPBEAT 2023

40%

of households in SSA lack reliable electricity access

STRUCTURAL WEAKNESSES IN SSA UTILITIES

Weak Revenue Collection

$5B+ annual losses. Manual billing and cash-based payments; collection rates often below 70%.

Aging Infrastructure and Reactive Maintenance

No real-time monitoring leads to costly unplanned outages and slow fault response.

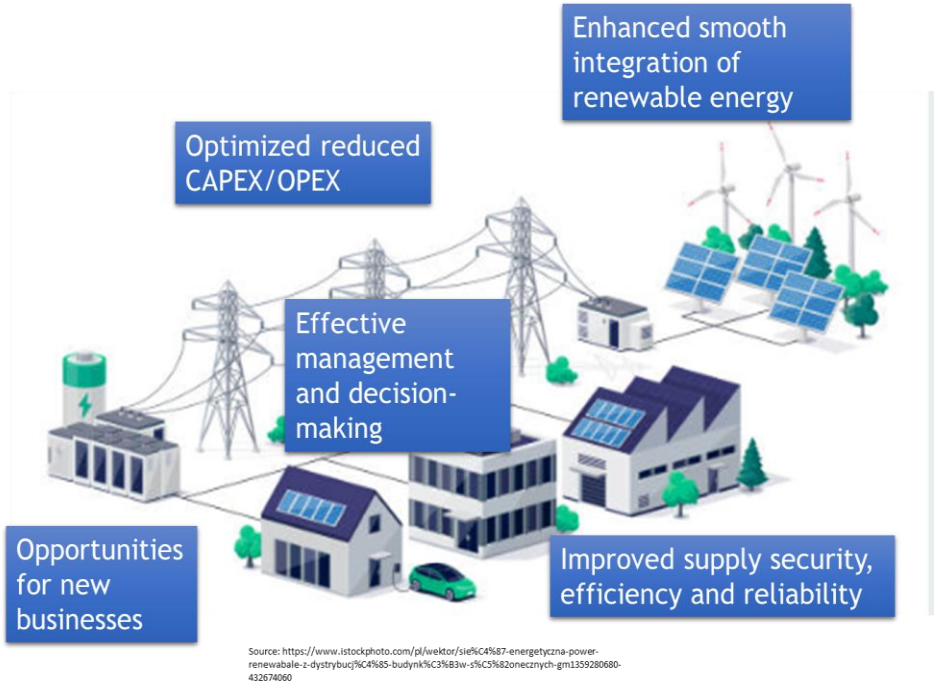
Skill Shortage

A lack of trained personnel restricts adoption and maintenance of advanced utility technologies and digitalization.

Sources: EMBER 2025; ev24.africa 2025; World Bank UPBEAT 2023; World Bank / IEA WDI

What Digitalization Delivers Across the Value Chain

Digital technologies improve efficiency at every stage — from generation to the customer meter



Source: https://www.istockphoto.com/pl/wektor/sie%C4%87-energetyczna-power-renewabale-z-dystrybucj%C4%85-budynk%C3%B3w-s%C5%82onecznych-gm1359280680-432674060

Optimized infrastructure use
Real-time monitoring can improve capacity use and defer investments

Improved Reliability
Real-time monitoring, predictive maintenance, faster fault restoration

Reduced T&D Losses
Smart metering + analytics cut both technical and commercial losses

Renewable Integration
Enables stable solar, wind and DER connections without grid instability

Consumer Engagement
Mobile-first billing, USSD, prepaid metering, self-service apps

New Business Models
VPPs, demand response, EV charging services, digital aggregation

IEA 2025 — Digitalization unlocks up to 175 GW of global transmission capacity

Source: IEA — Digitalisation and Energy (2025)

A Stylized Digital Journey for Utilities

Phased adoption of digital and smart grid technologies to address immediate priorities with a view towards future use-cases

Level 1. Traditional

- Manually Operated Distribution Grid
- Conventional Metering
- Low cyber readiness

Level 2. Developing

- Physical Network Transformation
- Traditional SCADA System
- Automated Meter Reading (AMR)
- Partial GIS

Level 3. Emerging

- Distribution and Substation Management
- Outage Management System (OMS)
- Initial Implementation of Advanced Metering (AMI)
- Partial system integration

Level 4. Advanced

- Smart Distribution and Substation Automation
- Full AMI implementation
- Enhanced SCADA
- Advanced Analytics
- Security Operations Center (SOC)
- Full scope GIS, system integration

Level 5. Intelligent

- Digital Twins
- DER participation
- DERMS
- Virtual Power Plants
- Digital Platforms

Increasing levels of visibility, granularity, control, and data-driven processes

Increasing levels of regulatory sophistication (tariff design, digital platforms, VPPs)

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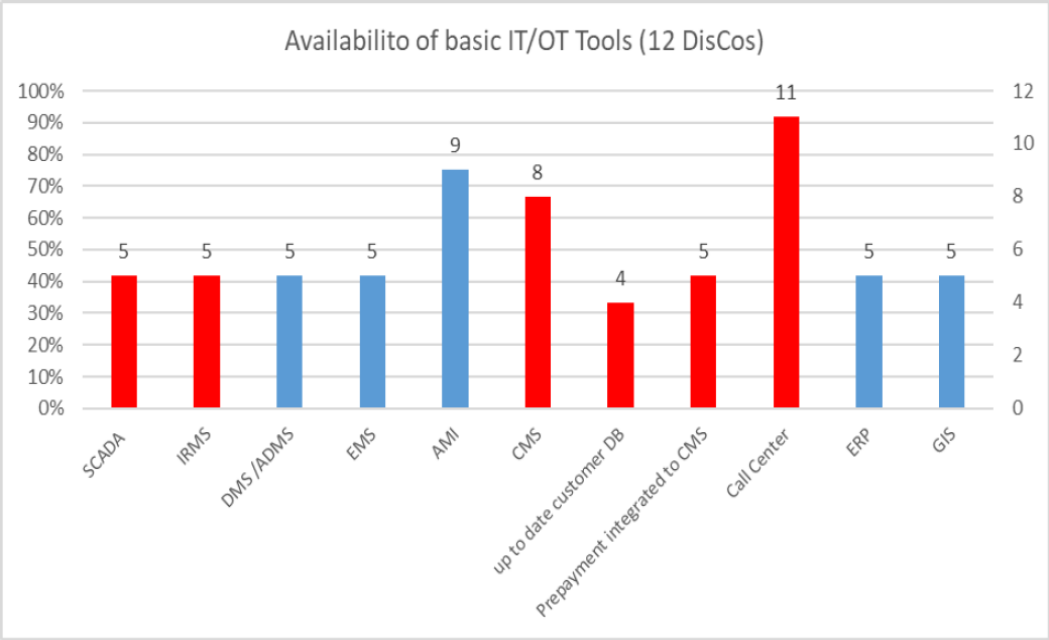
State of Digital Tools — West Africa Utility Snapshot

Assessment of 12 Distribution Companies (DisCos) in West Africa



KEY INSIGHTS

92%	Have a Call Center
75%	Have AMI (smart meters)
67%	Have CMS (billing system)
42%	Have SCADA, DMS, or EMS
33%	Up-to-date customer database

Most DisCos have customer-facing tools but lack the operational backbone (SCADA, data) needed for real efficiency gains.

Source: World Bank ESMAP assessment of 12 West Africa DisCos (2024)

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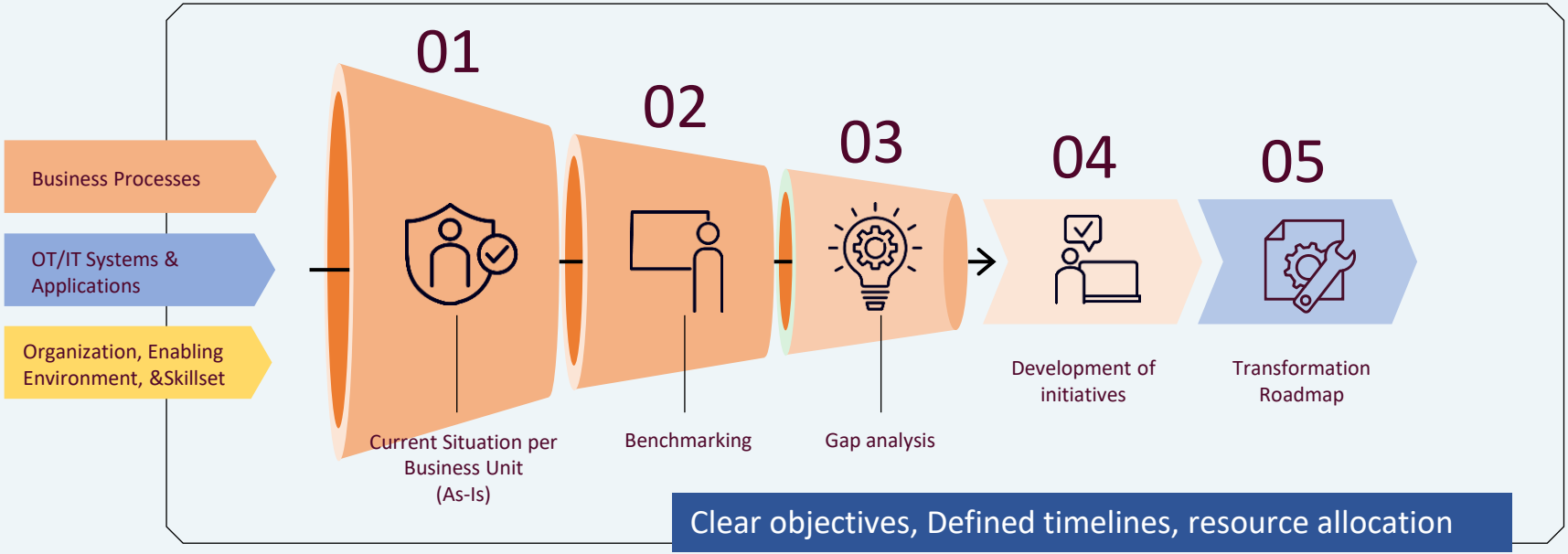
Illustrative Digital Roadmap Exercise



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Institutional Enablers

Governance, culture, leadership, and skills are critical

1.
Organizational Principles

A stable but adaptable operating model is needed to ensure accountability, collaboration, and keep pace with change

2.
Enabling Environment

Performance depends on leadership, policies, culture and partnerships that enable innovation and resilience

3.
Skillsets & Capabilities

Digitalization requires transversal skills that combine IT, data and operations to sustain agility and efficiency

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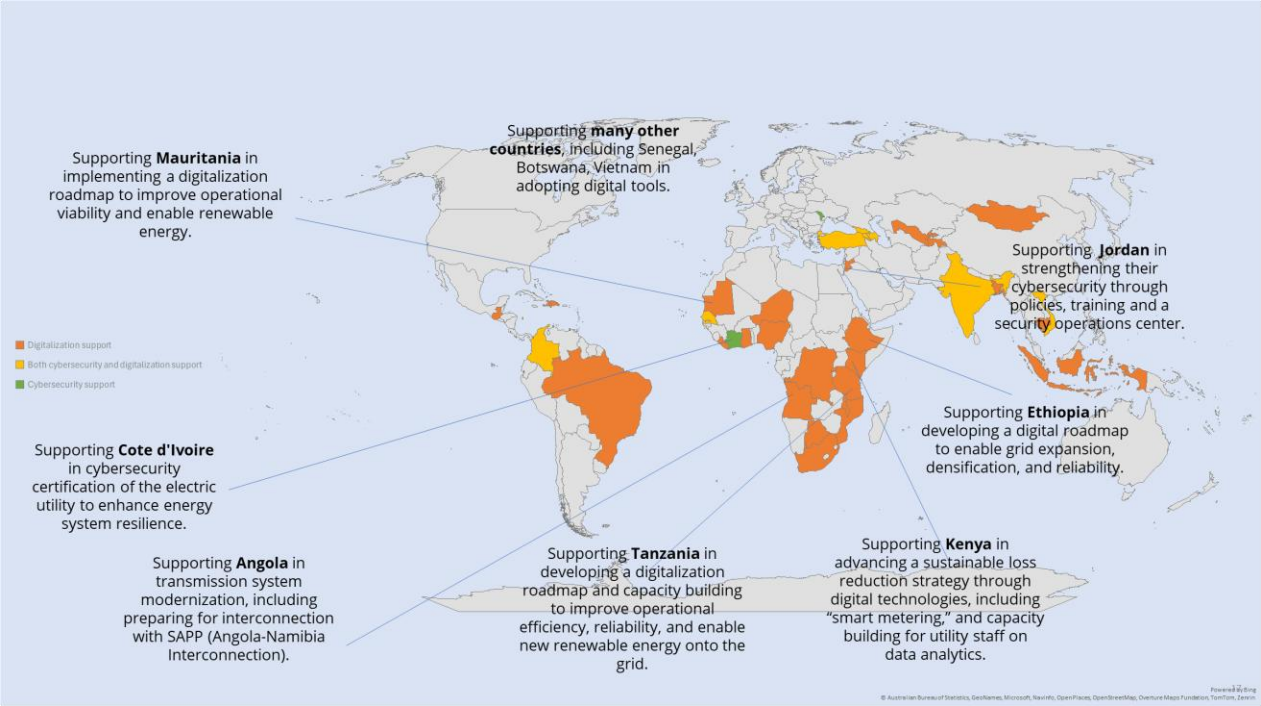
ESMAP Utility and Grid Modernization Program

Assistance from ESMAP

- Performance Improvement Plans
- Digital transformation roadmaps and implementation
- Enabling Distributed Energy Resources
- Cybersecurity Readiness
- Technical Assistance, Global Knowledge, and Knowledge Exchange
- Partnerships



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Capacity Development & Knowledge Exchange

Technology is not a panacea — success depends equally on human capital and institutional adaptation

Utility digitalization workshop — Africa

World Bank / ESMAP panel, Feb 2024

Smart Grid panel discussion — South Korea study tour

01

Peer-to-peer exchanges

Workshops, study tours and cross-utility partnerships

02

Technical training

Hands-on skills in IT/OT, data analytics & cybersecurity

03

South-South learning

Rwanda, Senegal, Tanzania sharing hard-won experience

Key Lessons– Challenges to Opportunity

1	Governance & leadership must come first — technology cannot fix institutional weaknesses
2	Embed training to build capacity and skills necessary for successful digital transformation
3	Treat data as an asset, value increases over time
4	Technology pilots to show proof of concept can unlock greater willingness and informed scale-up
5	Focus on employee upskilling and culture change